This is a slightly generalized form of the technique suggested by Bulgac and Kusnezov [58].

Assuming ρ goes to zero sufficiently rapidly at infinity, we have

$$\int \nabla_z \cdot \boldsymbol{G}(z)\rho d\omega = -\int (\nabla_z \rho)\cdot \boldsymbol{G}(z) d\omega = \beta \int \nabla_z H \cdot \boldsymbol{G}(z)\rho d\omega,$$

using integration by parts, therefore, since $\beta^{-1} = \mathrm{k_B T}$,

$$\mathrm{T} = \mathrm{k_B^{-1}} \frac{\langle \nabla_z H \cdot \boldsymbol{G}\rangle}{\langle \nabla \cdot \boldsymbol{G}\rangle},$$

which gives formulas for temperature in terms of phase space averages (compare Chap. 6, Proposition 6.1).

Suppose we let K be a scalar valued function of position, and set

$$\boldsymbol{G}(z) = \begin{bmatrix} \nabla_q K(\boldsymbol{q}) \\ \boldsymbol{0} \end{bmatrix}$$

then we arrive at

$$g(z) = \mu^{-1}[\nabla_q U - \mathrm{k_B T}\nabla_q] \cdot \nabla_q K(\boldsymbol{q}).$$

This is an example of a "configurational thermostat" [193, 211, 319].

A useful fact about thermostats is the following:

Proposition 8.1 (Thermostats are Additive) *Suppose we are given two thermostats,*

$$\dot{z} = \boldsymbol{J}\nabla H + \boldsymbol{\Phi}_1(\boldsymbol{\xi}, z), \qquad \dot{\boldsymbol{\xi}} = \boldsymbol{g}_1(\boldsymbol{\xi}, z),$$

with stationary density $\hat{\rho}_1(\boldsymbol{\xi})\rho_\beta$, and another

$$\dot{z} = \boldsymbol{J}\nabla H + \boldsymbol{\Phi}_2(\boldsymbol{\eta}, z), \qquad \dot{\boldsymbol{\eta}} = \boldsymbol{g}_2(\boldsymbol{\eta}, z),$$

with stationary density $\hat{\rho}_2(\boldsymbol{\eta})\rho_\beta$, then

$$\begin{aligned} \dot{z} &= \boldsymbol{J}\nabla H + \boldsymbol{\Phi}_1(\boldsymbol{\xi}, z) + \boldsymbol{\Phi}_2(\boldsymbol{\eta}, z), \\ \dot{\boldsymbol{\xi}} &= \boldsymbol{g}_1(\boldsymbol{\xi}, z), \\ \dot{\boldsymbol{\eta}} &= \boldsymbol{g}_2(\boldsymbol{\eta}, z) \end{aligned}$$

is another thermostat and has stationary density $\tilde{\rho} = \hat{\rho}_1(\boldsymbol{\xi})\hat{\rho}_2(\boldsymbol{\eta})\rho_\beta$.